

When Humans Adapted to Incomplete LLMs

Observational Record of External Cognitive Channel Formation in Early LLM Environments

Research Summary | Scope × COS Hypothesis Series | 2025

Overview

This paper documents an adaptive behavioral pattern observed on the human side during repeated high-abstraction, multi-domain interactions with early large language models (LLMs). It is not a completed theory, but an observational record — capturing how a user spontaneously restructured the human-AI interface in response to model limitations, and how those adaptations became visible precisely because of the friction that early systems produced.

Background & Observation Window

The observation originated in late 2024. During the transition to a newer model generation, the observer inadvertently continued using the legacy model (GPT 4-o series) for approximately six months. Upon switching, a mismatch between the previously formed interaction patterns and the new model's response tendencies created an unintended comparative window — making output-level differentiation behaviors visible that would otherwise have gone unnoticed.

This accidental pathway is itself methodologically significant: in current high-performance LLMs, the instabilities that enabled this observation have largely been smoothed over. The window has closed.

Core Finding: Three-Channel Segmentation

Because a single conversational channel proved insufficient for sustaining high-abstraction dialogue, the user externally designed three purpose-specific channels — each operating the same underlying model under different prompt configurations:

Channel	Function	Primary Use
High-Abstraction Mode	Cross-domain structural reasoning	PDF production, complex multi-layered dialogue
Structural Mode	Knowledge explanation & strategy	Clarification, intermediate translation
Empathic Mode	Emotional stabilization	High-load states, conversational continuity

Importantly, this segmentation was not intentional from the outset — it emerged naturally from repeated attempts to resolve interaction friction, and was only recognized as a three-channel structure in retrospect.

Why This Adaptation Emerged

Model-side constraints:

- Empathy-oriented processing dominated structural responses, reducing precision for logical inputs.
- Consistency degraded under high-abstraction, multi-layered inputs.

User-side conditions:

- A cognitive tendency to extract structural patterns from everyday concrete observations, then inject them into dialogue at high abstraction levels.
- Consistent demand for structural coherence over emotional alignment.
- Layered input noise: Japanese linguistic ambiguity compounded by 2E (Twice-Exceptional) typing patterns, creating an unusually demanding input environment.

Comparative Observability Across Models

The three-channel structure was only observable within the legacy GPT environment. This was not because other models were inferior — it was a question of observational conditions:

- **Legacy GPT:** Mismatches surfaced readily, triggering visible adaptive behavior.
- **Gemini:** Friction was absorbed through empathic supplementation — mismatches rarely triggered adaptation.
- **Claude:** Structural follow-through from the outset reduced friction below the threshold needed to trigger adaptive behavior.

The paradox: model imperfection functioned as a lens. As models improve, human adaptive structures become internalized and invisible. This paper records a window that is now closing.

HCI Interpretation

From a Human-Computer Interaction perspective, this case represents a user restructuring the interface according to their own cognitive processing requirements — not as a designed experiment, but as emergent behavior under operational pressure.

Framed through Extended Cognition theory, the three channels functioned as external implementations of self-observation, structuring, and emotional-processing support. Notably, the processing vector was directed inward: emotional states were not resolved through empathic exchange, but by structuring why they occurred — making reproducibility and countermeasures visible.

Conclusion

When humans adapt to incomplete systems, they may externalize cognitive structures. The adaptive pattern described here was not designed — it precipitated from friction. As LLMs become increasingly integrated and high-performing, such adaptive traces will become unobservable. This paper is positioned as a record from the transitional period in which that observational window was still open.

Full operational logs and dialogue records available in: Scope × COS Operational Verification Log v3